

EMMA LINSComb

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EDUCATION

Cornell University, Ithaca, NY

B.A., Physics, 2023 to 2027 (expc.)

Thesis Title: Examining Tidal Deformability of Neutron Stars by Simulating Inspiral Degeneracy of Differing Internal Structures

Select Coursework: Introduction to Computing: Design and Development in Python • Waves and Thermodynamics • Analytical Mechanics • Advanced Electricity and Magnetism • Intermediate Quantum Mechanics • Introduction to Particle Physics • Multiwavelength Astronomical Techniques • Multivariable Calculus • Differential Equations • Linear Algebra • Applied Complex Analysis • Partial Differential Equations • Science in American Politics

GPA: 3.91/4.3

RESEARCH EXPERIENCE

Professor Nils Deppe Computational Relativity Research Group

Cornell University

Research Assistant

Aug 2025 - Present

- Implemented modified Tolman–Oppenheimer–Volkoff Equations in Python to numerically solve theoretical neutron stars with different equations of state (EOS).
- Performed rigorous error analysis to determine the validity of simulated solutions.
- Wrote C++ code that contributed to the SpECTRE project and improved the accuracy of neutron star solutions for polytropic EOS, using a rigorous expansion near the stellar center.
- Investigated the waveforms of neutron star binaries with degenerate tidal deformabilities, but different internal EOS.

Dr. Amanda Bauer Observational Astronomy Research Group

Yerkes Observatory

Research Assistant

May 2025 - Present

- Examined the growth of the Dumbbell Planetary Nebula (M27) over a period of 119 years.
- Operated the Yerkes 41" Reflector for data collection and the Yerkes 40" Great Refractor for optical observations.
- Handled, digitized, and inspected historical glass plate observations from the Yerkes 180,000 glass plate collection.
- Utilized astropy, photutils, and publicly available astronomy programs such as SAO DS9 and AstroImageJ to compare historical glass plate observations to modern-day digital observations.
- Presented final findings in a 15-minute presentation to the Yerkes board of directors, and wrote a final paper documenting methods and results.

Professor Abigail Crites Observational Cosmology Research Group

Cornell University

Research Assistant

Jan 2024 - Present

- Examined [CII] emission spectra from the early universe.
- Implemented code for the calibration of the emission spectra detectors.
- Installed emission spectra detectors and tested wiring throughout the instrument.
- Operated heavy machinery to assemble and disassemble the instrument.
- Collaborated with a team of 4 other undergrads on mechanical issues in the instrument.
- Presented final results to an audience of my peers during the Cornell Nexus Scholars summer program.

AWARDS AND FELLOWSHIPS

Cornell Arts & Sciences Nexus Scholar (\$7,000)	2024
<i>Fellowship funding a 9-week, full-time summer research experience at Cornell</i>	
SPS Association for Women in Science Kirsten R. Lorentzen Award (\$2,500)	2026
<i>This award is given annually to an exceptionally well-rounded student who excels in her studies as well as outdoor activities, service, sports, music, or other non-academic pursuits or who has overcome significant obstacles.</i>	

PRESENTATIONS

247th American Astronomical Society Meeting	01/2026
Poster: <i>Examining the Growth of the Dumbbell Nebula, M27, with 119 Years of Observations</i>	
Cornell Undergraduate Physics Poster Session	12/2025
Poster: <i>Examining the Growth of the Dumbbell Nebula, M27, with 119 Years of Observations</i>	
Yerkes Board of Directors Colloquium	08/2025
Talk: <i>Measuring Growth of the Dumbbell Nebula</i>	
Cornell Nexus Scholars Capstone	07/2024
Talk: <i>Optimization and Lab Characterization of the TIME Instrument</i>	

MENTORSHIP

Cornell Physics Department Learning Assistant Program	Cornell University
<i>Learning Assistant</i>	01/2026 - Present
• Attended weekly classes focused on developing teaching skills. • Worked with the course team during weekly meetings to address the content of the course and the performance of the students. • Spent ~ 2 hours per week running office hours and assisting students in developing their conceptual understanding of the course content.	
Society of Physics Students Peer Mentorship Program	Cornell University
<i>Peer Mentor</i>	08/2024 - Present
• Mentored 7 students total: 3 in the 2024-2025 academic year, 4 in the 2025-2026 academic year. • Met with freshmen mentees once a month and guided them through the process of adjusting to the physics major and college life in general.	

SCIENCE LEADERSHIP AND SERVICE

Cornell Society of Physics Students	Cornell University
<i>President</i>	12/2025 - Present
• Managed about a dozen events per semester, including speaker events, career panels, socials, mentorship programs, local outreach, the Cornell Physics Wiki, and an annual poster session. • Served on the CEDI committee, and continue to facilitate community-building initiatives in the Cornell Physics Department (more information below).	

- Organized the initiatives of the organization, promoting professional development and community.
- Planned the third-annual AEPxPhysics Formal in collaboration with the Applied Engineering Physics Society and the Physics Graduate Society, taking on the primary leadership role for the planning committee.

Vice President

12/2024 - 12/2025

- Organized and oversee ~ 10 physics events per semester, collaborating with the president and executive board.
- Led our undergraduate peer mentorship program of ~ 100 participants and our undergraduate-graduate mentorship program with ~ 12 participants.
- Served on the CEDI committee and complete the “A Physicist Is” project for the second cohort (more information below).
- Planned and hosted the second-annual AEPxPhysics Formal, working with the Applied Engineering Physics Society.
- Wrote an article for the SPS Observer, featured in the Winter 2025 issue.

Women in Physics Chair

02/2024 - 12/2024

- Planned and host ~ 10 physics events per semester: including professional development workshops, speakers, socials, etc.
- Served as a student-faculty liaison on the Cornell Physics Committee for Climate, Equity, Diversity, and Inclusion (CEDI).
- Developed “A Physicist Is” under Prof. Holmes, a project to showcase the diversity of careers and identities in physics through informational graphics posted around the Physical Sciences Building.

OTHER LEADERSHIP AND SERVICE

Hearsay A Cappella

Cornell University

Alumni Chair

5/2026 - Present

- Kept in constant contact with Hearsay alumni to facilitate travel to future concerts, get donations, and support community efforts.

Choreography Chair

5/2025 - Present

- Designed choreography for all songs in the set-list, and led dancing-focused rehearsals for 15 members.
- Worked closely with the music director and president to schedule and plan extra rehearsals in the weeks leading up to the concert.
- Held weekly office hours for members needing help with their choreography.
- Filmed practice runs of the show and wrote individualized notes for every member based on their performance.

Assistant Music Director

5/2025 - Present

- Arranged multiple pieces for performance and taught newer members how to read and write music.
- Led singing-focused rehearsals for 15 members when the main music director was absent, and assisted the main music director with leading warm-ups and giving performance feedback.

- Coordinated group auditions with the group's president, business manager, and music director.
- Held weekly office hours for members needing help learning their music or preparing for their solos.
- Taught members how to beatbox.

Internal Social Chair

5/2024 - 12/2024

- Planned and hosted social events for members throughout the semester, including new member celebration, senior sendoff, and mid-semester date night.
- Organized the big-little program.

SKILLS

Programming: Python (NumPy, SciPy, Matplotlib, pandas), C++, Git, Bash, L^AT_EX, Windows, MacOS

Scientific Computing: Numerical Methods, Error Analysis, Scientific Computing, Computational Physics, Data Analysis

Astrophysics Software: AstroPy, MaximDL, AstroImageJ, SAODS9, Siril

Experimental & Observational: Telescope Operation, Detector Calibration, Scientific Instrumentation, Archival Astronomical Data, Image Reduction, Instrument Assembly

Research: Scientific Writing, Literature Review, Technical Presentations, Collaborative Research

Language: Native Fluency in English, B2 Fluency in French

Management: Teaching, Mentorship, Budget Management, Leadership, Large Event Planning, Delegation

AFFILIATIONS

Member, American Astronomical Society

Member, American Physical Society

Member, Society of Physics Students